
Yanir Kleiman

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Experienced generative AI researcher with a strong background in computer graphics, animation, and VFX. I am currently a senior research scientist at Kyoso, previously at Meta AI (FAIR / GenAI).

I completed my PhD at Tel Aviv University, working on shape analysis, segmentation and shape matching. In recent years I have led research projects in generative 3D modeling, image and video generation, diffusion models, LLM / VLM training, and conversational agents.

I have prior in-depth practical experience in animation and VFX production.

Professional Experience

2025 - now

Senior Research Scientist, Kyoso

Kyoso is a series A startup developing an AI creative agent for graphic design, motion graphics, and animation.

- Leading the animation and motion graphics research efforts.
- 0-to-1 development of an animation agent, including data collection, training, and evaluation.
- Combining agent skills with fine-tuned LLMs to produce animation plans, keyframe data, and video editing.
- Ownership of animation features, coordinating engineering and design teams to integrate research into user-facing product.

2020 - 2025

Research Engineer, Meta AI

Llama team:

- Llama 3 and Llama 4 multi-modal pre-training and post-training
- Procedural generation of training data for visual math problems
- Augmenting and improving training data with LLM/VLM as a judge
- Directly contributed to increase in model performance over visual reasoning benchmarks (MMMU, MathVista)

Selected projects at FAIR / GenAI:

- Meta 3D TextureGen (lead researcher, joint first author)
- Meta 3D AssetGen
- Replay dataset: aligned multi-modal multi-view videos for 3D reconstruction (joint first author)

Technical leadership of 3D shapes dataset management and curation:

- Evaluation and quality estimation of massive datasets from external vendors
- Ingestion and conversion of 2M shapes in various 3D formats
- Shape preprocessing and rendering setups
- PBR materials manipulation and verification
- Extracted millions of auxiliary assets (textures, materials, renders) for training purposes.

Contributions to Spark AR - a 3D studio software that artists use to create effects for mobile phones and headsets:

- Incorporated real time location tracking simulation in Spark Studio
- Integrated simulated 3D environments into Spark Studio

2019 - 2020

Research Scientist, Shapes AI

An early stage startup company that developed solutions in the space of 3D shapes and video-based visual reasoning. I led the research efforts in the 3D domain, which include generative modeling, point clouds processing, and 3D search.

- Developed a novel patch-based deep learning method to clean up noisy point clouds.
- Developed an end-to-end solution for generating thousands of textured shapes from a small set, which includes procedural modeling and texture generation.
- Delivered a set of 50K objects to a customer for robotics software training.
- Worked on 3D shapes search where the input is an image or a 3D shape.

2018 - 2019	Software Developer, DNEG DNEG is a high-end visual effects company. I developed tools for the "on-set" department which manages and curates the reference footage on the set of a show. • Developed and supported tools for browsing massive sets of 100K+ images which are used for every production in the company. • Developed a deep learning method to detect color charts in images, automating a previously manual process. • Developed a novel deep learning method to remove noise artifacts and increase quality of compressed source material (see 2019 publication). • Worked with Maya, Nuke, Alembic, libraw, OpenEXR, and other common tools of the VFX industry, using mostly C++, Python, and MEL scripts.
2016 - 2017	Post-doc Researcher, Laboratoire d'Informatique, École Polytechnique, France. Part of a computer graphics group led by Maks Ovsjanikov.
2011 - 2016	PhD Student, Computer Science, Tel Aviv University, Israel. Computer graphics lab, under the supervision of Prof. Daniel Cohen-Or.
2008-2010	3D Animation / VFX Artist, Various Projects Lighting, shading, rendering, visual effects, compositing, tracking, scripting. • Crew 972 (for Warner Bros.) Looney Tunes Show, Roadrunner segments • Gravity Israel Visual Effects Flagship broadcast commercials • "Deus" (sci-fi TV show) Visual effects shots over live plates • Vancouver Film School 3D Animation and Visual Effects Diploma, graduated with honors
2008	Algorithms Developer, MutualArt Inc. Developed and implemented automated text categorization and linking algorithms.
2005 - 2007	Engineering Manager, Amobee Media Systems Ltd. Led a team of 5 engineers in an early stage ad-tech startup. • People management, task planning, recruiting and training. • .NET infrastructures (ASP.NET), Java, and SQL Server development.

Expertise

ML / AI	Experience with diffusion models, flow matching, LLMs, VLMs, and agentic workflows.
Computer Vision	Image and video generation, video understanding, optical flow, object detection, segmentation, tracking, structure-from-motion, multi-view reconstruction.
3D Geometry	Working with meshes, point clouds, and implicit 3D representations (NeRF, SDF, GS).
Media Production	Deep knowledge of content production pipelines, including advanced scripting in Blender, Maya, Nuke, and After Effects.

Academics

2016	PhD in Computer Science, Tel Aviv University, Israel. <i>Dissertation:</i> Semantic Similarity and Correspondence of 3D Shapes and Images.
2005	M.Sc. in Computer Science, Tel Aviv Academic College, Israel Graduated with honors.
2000	B.Sc. in Math and Computer Science, Tel Aviv University, Israel

Publications

*Joint first authors marked with **

2025	InstanceGen: Image Generation with Instance-level Instructions Etai Sella, Yanir Kleiman, Hadar Averbuch-Elor SIGGRAPH, 2025
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- 2024 **The Llama 3 herd of models**
Abhimanyu Dubey et al. (500+ authors)
arXiv, 2024
- 2024 **Meta 3D Gen (technical report)**
*Raphael Bensadoun, *Tom Monnier, *Yanir Kleiman, Filippos Kokkinos, Yawar Siddiqui, Mahendra Kariya, Omri Harosh, Roman Shapovalov, Benjamin Graham, Emilien Garreau, Animesh Karnewar, Ang Cao, Idan Azuri, Iurii Makarov, Eric-Tuan Le, Antoine Toisoul, David Novotny, Oran Gafni, Natalia Neverova, Andrea Vedaldi
Meta Research, 2024
- 2024 **Meta 3D AssetGen: Text-to-mesh generation with high-quality geometry, texture, and pbr materials**
Yawar Siddiqui, Tom Monnier, Filippos Kokkinos, Mahendra Kariya, Yanir Kleiman, Emilien Garreau, Oran Gafni, Natalia Neverova, Andrea Vedaldi, Roman Shapovalov, David Novotny
NeurIPS, 2024
- 2024 **Meta 3D TextureGen: fast and consistent texture generation for 3d objects**
*Raphael Bensadoun, *Yanir Kleiman, Idan Azuri, Omri Harosh, Andrea Vedaldi, Natalia Neverova, Oran Gafni
arXiv, 2024
- 2023 **Replay: Multi-modal multi-view acted videos for casual holography**
*Roman Shapovalov, *Yanir Kleiman, *Ignacio Rocco, David Novotny, Andrea Vedaldi, Changan Chen, Filippos Kokkinos, Ben Graham, Natalia Neverova
ICCV, 2023
- 2019 **Boosting VFX Production with Deep Learning**
Yanir Kleiman, Simon Pabst, Patrick Nagle
ACM SIGGRAPH Talks, 2019
- 2018 **PCPNet: Learning Local Shape Properties from Raw Point Clouds**
*Paul Guerrero, *Yanir Kleiman, Maks Ovsjanikov, Niloy J. Mitra
Eurographics, 2018
- 2018 **Group optimization for multi-attribute visual embedding**
Qiong Zeng, Wenzheng Chen, Zhuo Han, Mingyi Shi, Yanir Kleiman, Daniel Cohen-Or, Baoquan Chen, Yangyan Li
Visual Informatics, 2018
- 2018 **Dance to the beat: Synchronizing motion to audio**
Rachele Bellini, Yanir Kleiman, Daniel Cohen-Or
Computational Visual Media, 2018
- 2018 **Robust Structure-Based Shape Correspondence**
Yanir Kleiman, Maks Ovsjanikov
Computer Graphics Forum, 2018
- 2017 **Region-Based Correspondence Between 3D Shapes via Spatially Smooth Biclustering**
Matteo Denitto, Simone Melzi, Manuele Bicego, Umberto Castellani, Alessandro Farinelli, Mario A. T. Figueiredo, Yanir Kleiman, Maks Ovsjanikov
ICCV 2017
- 2016 **Semantic Similarity and Correspondence of 3D Shapes and Images**
Yanir Kleiman
PhD Dissertation
- 2016 **Time-varying Weathering in Texture Space**
Rachele Bellini, Yanir Kleiman, Daniel Cohen-Or.
SIGGRAPH, 2016
- 2016 **Toward Semantic Image Similarity from Crowdsourced Clustering**
Yanir Kleiman, George Goldberg, Yael Amsterdamer, Daniel Cohen-Or.
CGI, 2016

- 2015 **SHED: Shape Edit Distance for Fine-grained Shape Similarity**
Yanir Kleiman, Oliver van Kaick, Olga Sorkine-Hornung, Daniel Cohen-Or.
SIGGRAPH Asia, 2015
- 2015 **DynamicMaps: Similarity-based Browsing through a Massive Set of Images**
Yanir Kleiman, Dov Danon, Jasmin Felberbaum, Joel Lanir, Daniel Cohen-Or.
CHI, 2015
- 2014 **Shape Segmentation by Approximate Convexity Analysis**
Oliver van Kaick, Noa Fish, Yanir Kleiman, Shmuel Asafi, Daniel Cohen-Or.
ACM Transactions on Graphics (TOG), 2014
- 2013 **Dynamic Maps for Exploring and Browsing Shapes**
Yanir Kleiman, Noa Fish, Joel Lanir, Daniel Cohen-Or.
SGP, 2013
- 2011 **Unsupervised co-segmentation of a set of shapes via descriptor-space spectral clustering**
Oana Sidi, Oliver van Kaick, Yanir Kleiman, Hao Zhang, Daniel Cohen-Or.
SIGGRAPH Asia, 2011
- 2007 **Paging with connections: FIFO strikes again**
Leah Epstein, Yanir Kleiman, Jiri Sgall, Rob van Stee.
Theoretical computer science, 2007

Awards and Fellowships

- 2016-2017 **Chateaubriand Fellowship for Postdoctoral Research**
- 2014-2015 **Google Focused Research Award**
I was partly funded by this grant during my PhD studies.
- 2010 **Animex Visual Effects Award**
Runner up for Best Visual Effects in Animex 2010 Festival.
- 2005 **Excellence Scholarship - Tel Aviv Academic College**
Awarded for excellence during my M.Sc. studies.